

2025

Dream House Design Project

MINIMALIST ARCHITECTURE
CONNOR LANGSFORD

HAMILTON SOUTHEASTERN HIGHSCHOOL

Dream Home Design

Connor Langsford

Gish

Civil Engineering and Architecture

3/5/2025

Abstract

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Preliminary Research

-Frank Lloyd Wright

Frank Lloyd Wright had a career that lasted seven decades and designed 1,114 architectural works. He incorporated many types of architectural styles, but most of his famous works, like Fallingwater, became known as the prairie style. It featured a long, low, and open floor plan that emphasized the horizontal line.

Falling Water



Frank Lloyd Wright, 1939

Jacobs House



Frank Lloyd Wright, 1937

House	Falling Water	Jacobs House
Architectural Style	-Organic architecture -Modernist -Prairie	-Modernist -Prairie
Architectural Elements	-Integration of nature -Horizontal lines -Open floor plan	-Horizontal lines -Low ceiling -Affordable design

Final architectural concept Design

Minimalism: Simple elegant design that focuses on clean lines and open spaces. The use of simple geometric shapes and natural color schemes creates a harmony and calmness to the overall architectural design.

Criteria and Constraints

-Architectural Design

- The architectural design of the shelter shall be complementary to the residential character of the development and be supplemented with landscaping
- All siding shall be masonry, wood, fiber cement board siding, stucco, composite lap siding, decorative precast panels, aluminum, EIFS, or heavy-gauge vinyl (minimum of 0.042" gauge). 3Lap siding shall have a maximum ten-inch (10") exposed board face. Vinyl siding shall be nailed and not stapled to the side of the house.

Site Information

17701 W Glenn Hwy, Sutton, AK 99674 (Sheep Mountain Lodge)



Code Requirements

-Zoning

The R1 (Residential) District is intended to provide for single-family detached homes in a low-density range of approximately 1.5 dwelling units per acre

-Structure Height

2.5 stories or 35 feet, whichever is greater

Preliminary Design

Standard Subdivision Development and Design Standards								
		Zoning District						
		ER	R1	R2	R3	R4	R5	M1
Lot Standards	Min. Lot Area	34,848	16,000	12,000	9,600	8,000	6,000	1,500
	Min. Average Lot Area	43,560	20,000	15,000	12,000	10,000	7,500	1,500
	Min. Lot Frontage	50	15	15	30	50	50	20
	Min. Lot Width at Building Line	150	125	100	90	80	70	20
Setback Standards	Min. Front Setback	75	50	40 (30')	30	25	25	15
	Min. Side Setback	30	20	15;10'	10; 5'	10	10	0
	Min. Aggregate Side Setback	n/a	n/a	30	20	n/a	n/a	n/a
	Min. Rear Setback	50	50	30	30	25	25	18
	Min. Building Separation	n/a	n/a	n/a	n/a	n/a	n/a	25
Other Standards	Max. Impervious Surface Coverage	35%	35%	35%	40%	40%	45%	65%
	Min. Living Unit Area	2,400	2,200	2,000	1,800	1,600	1,400	n/a
	Min. Average Living Unit Area	2,600	2,400	2,200	2,000	1,800	1,600	1,200
	Base Density (d.u./Ac.)	n/a	1.5	1.7	2.0	2.5	3.0	5.0
	Min. Base Open Space	0% ³	5% ³	10% ³	18%	18%	18%	18%

Viability Analysis

☐ Is the project legally permissible?

Yes, passes all specified code requirements

☐ Is the project physically possible?

yes

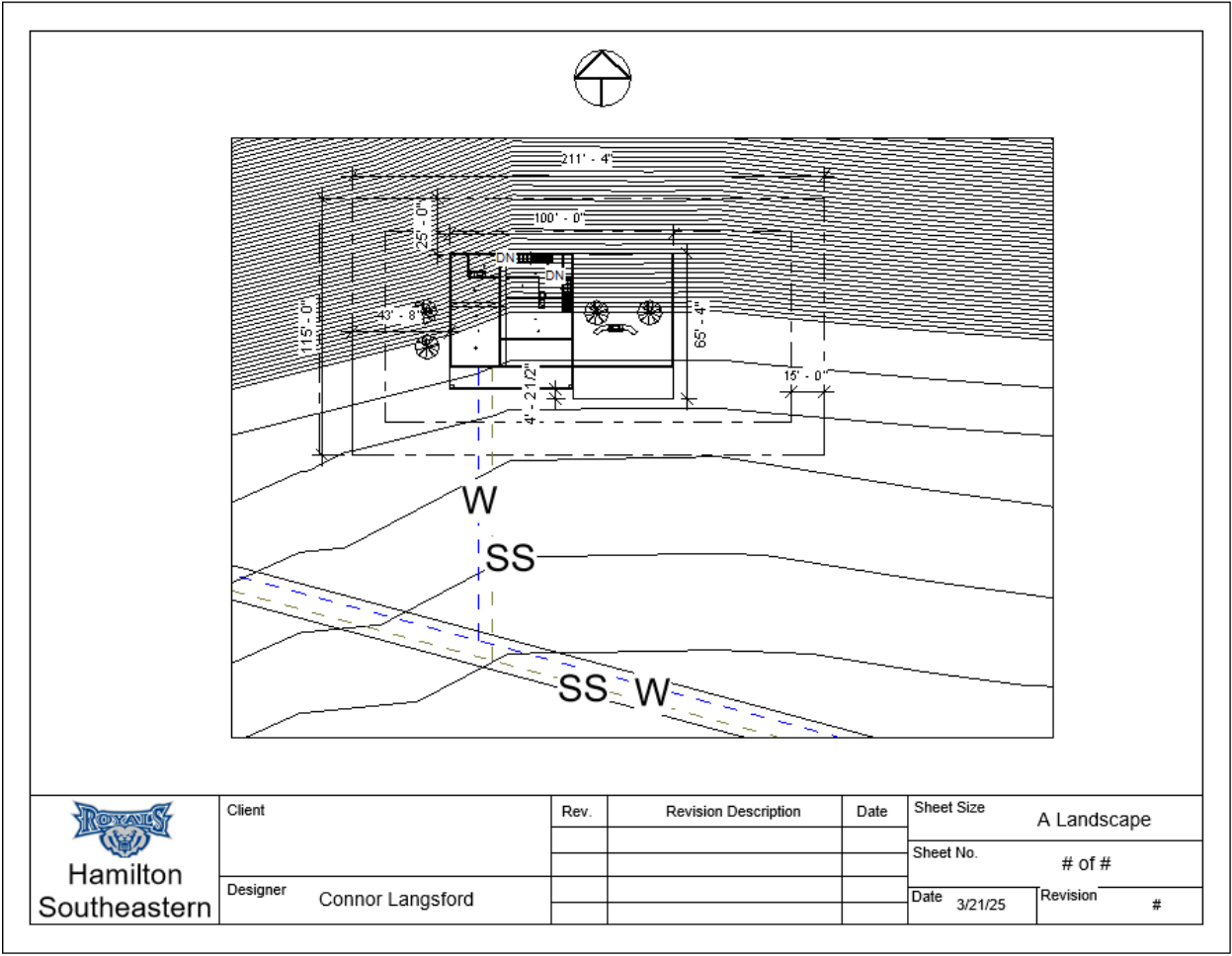
☐ Is the project financially feasible?

yes

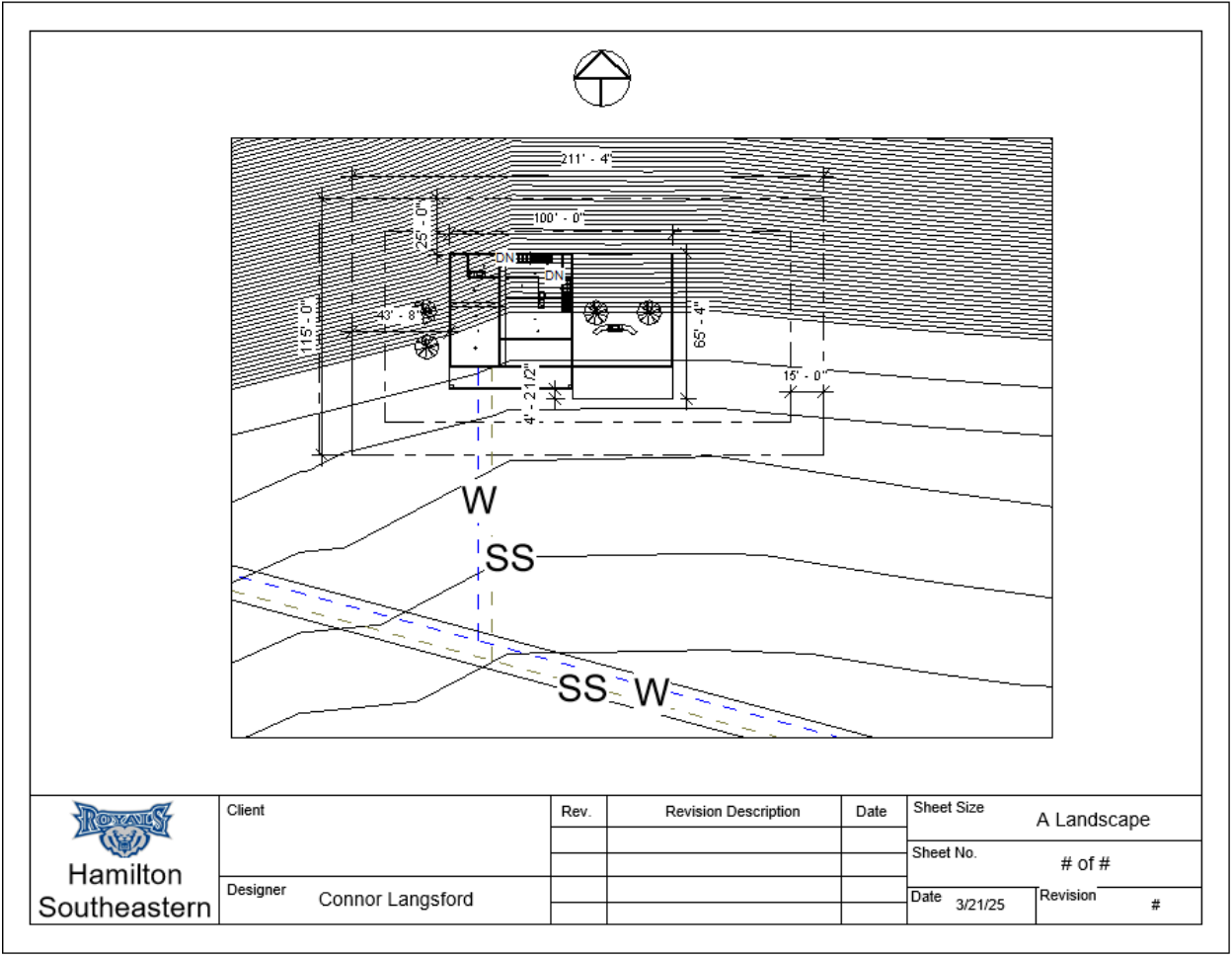
☐ Is the project a good choice for the community?

yes

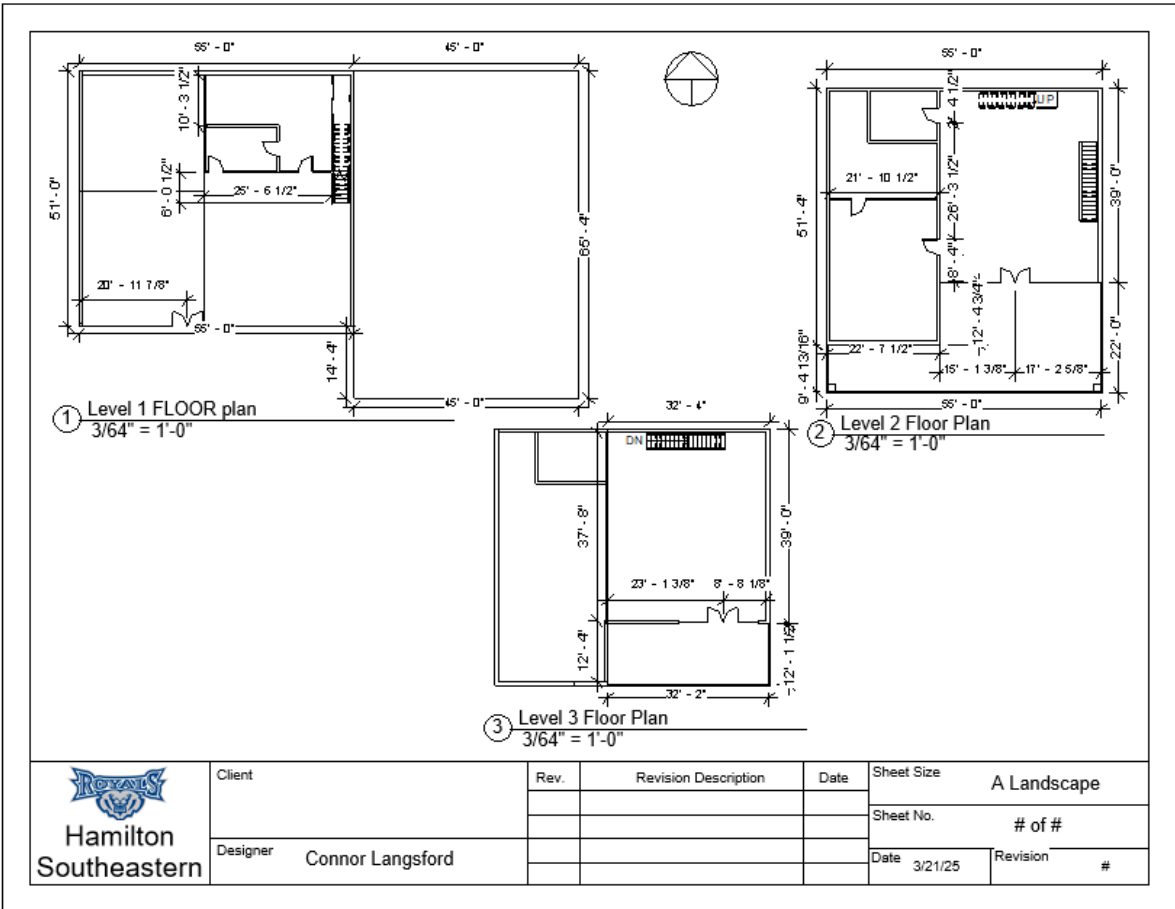
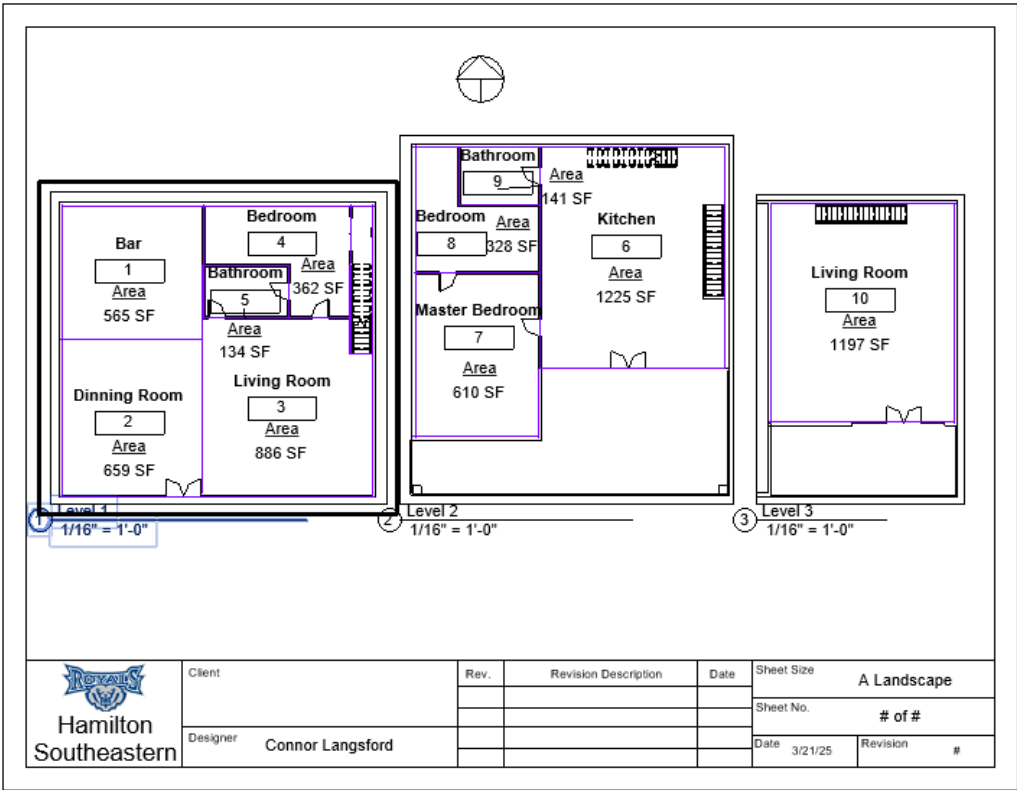
Site/Landscaping Plan



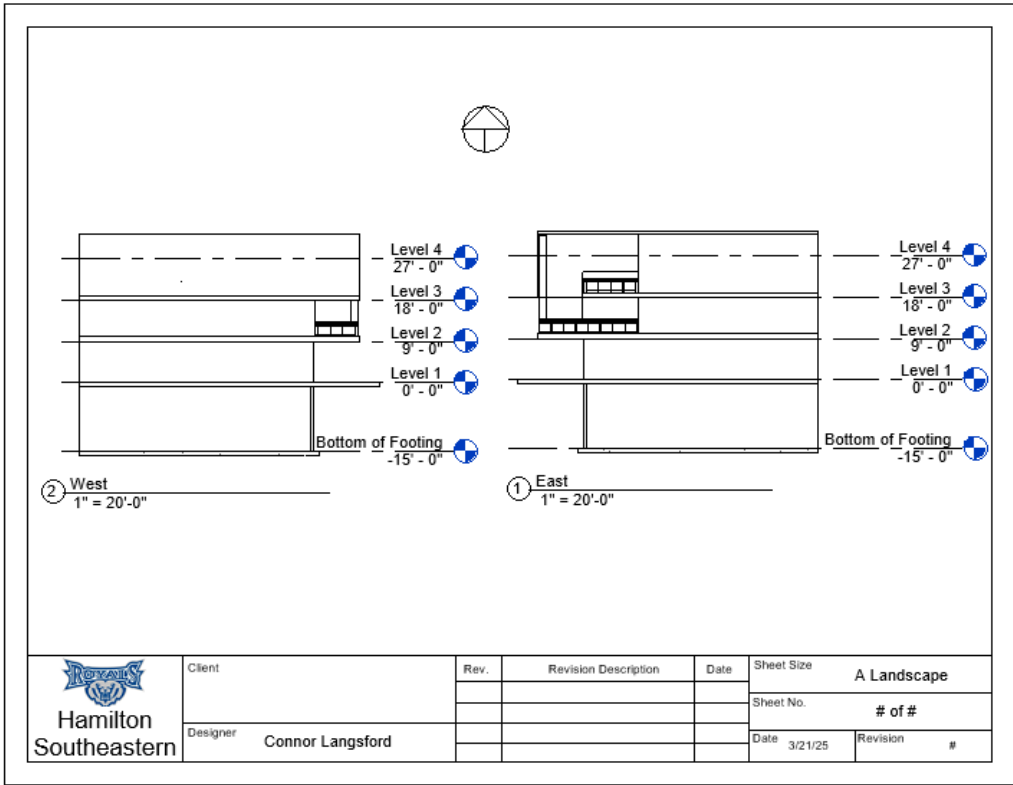
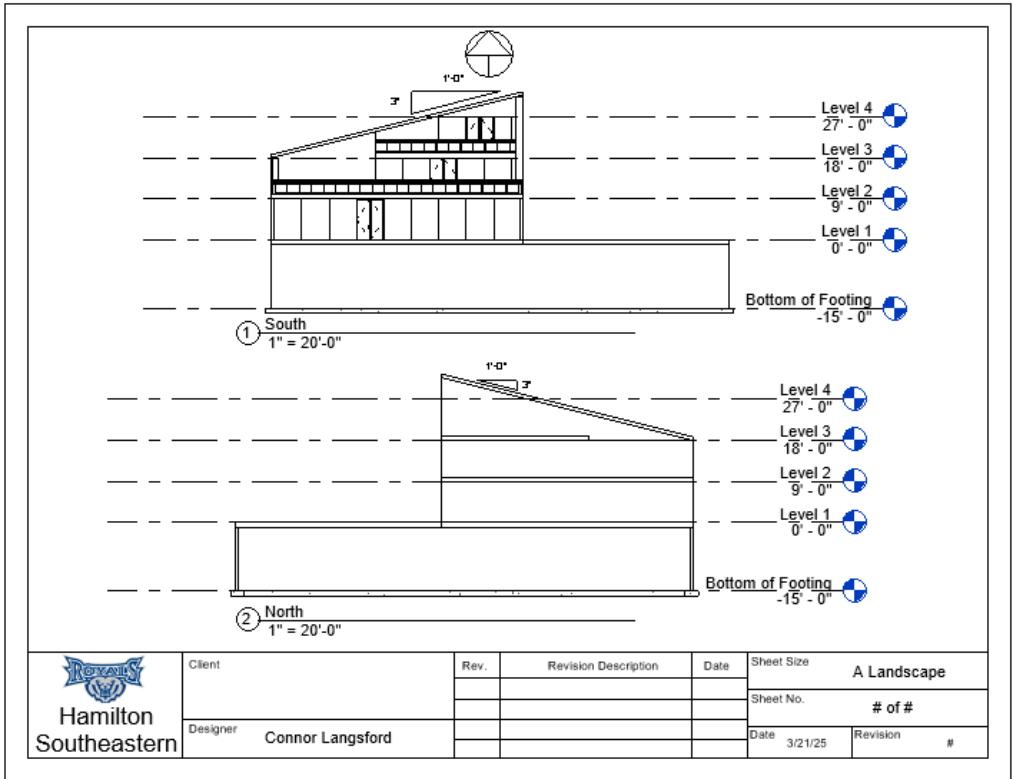
Foundation Plan



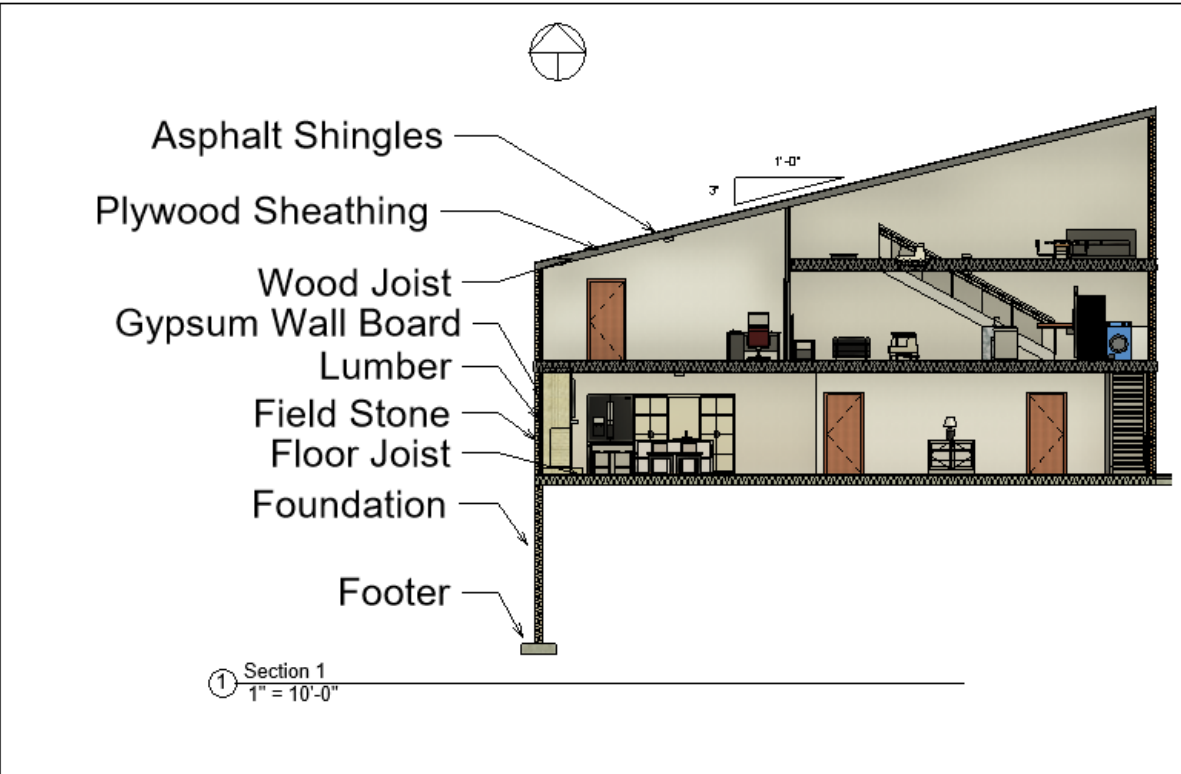
Floor Plans



Elevations

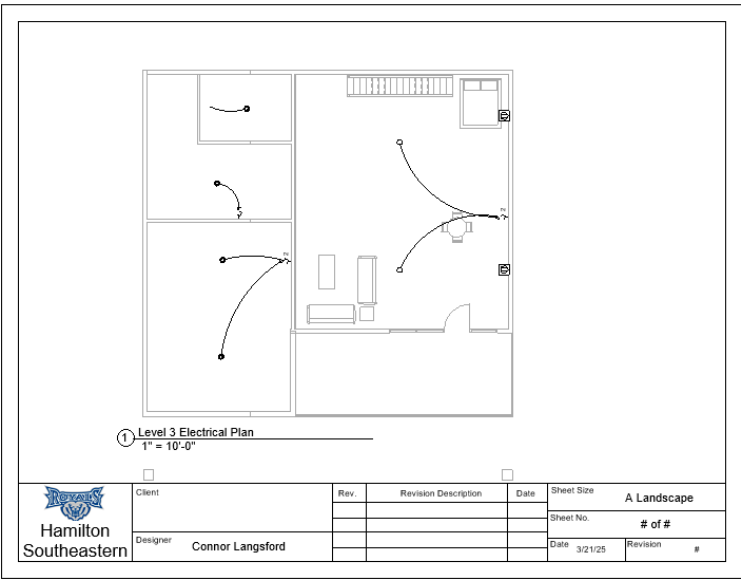
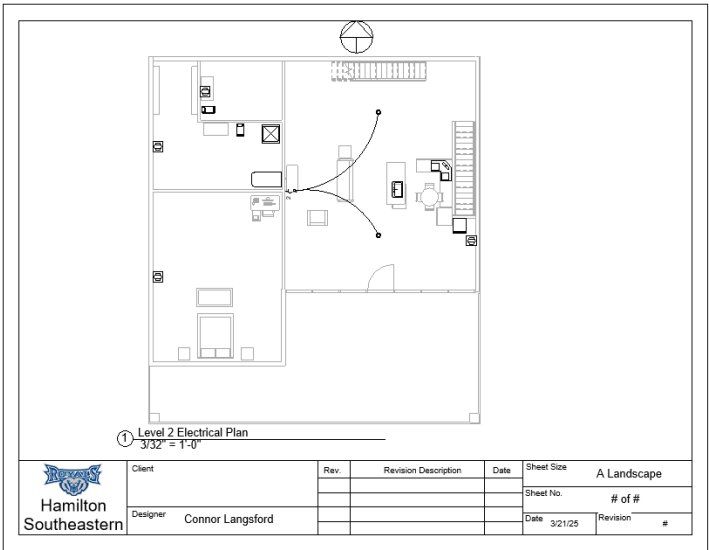
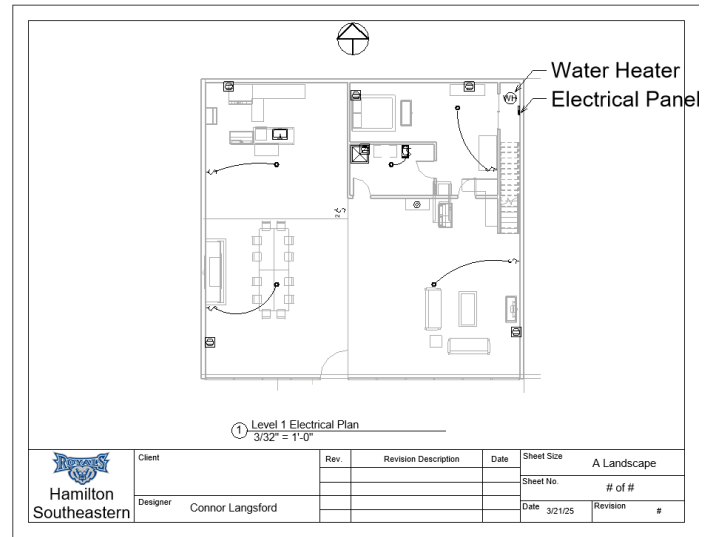


Section Views



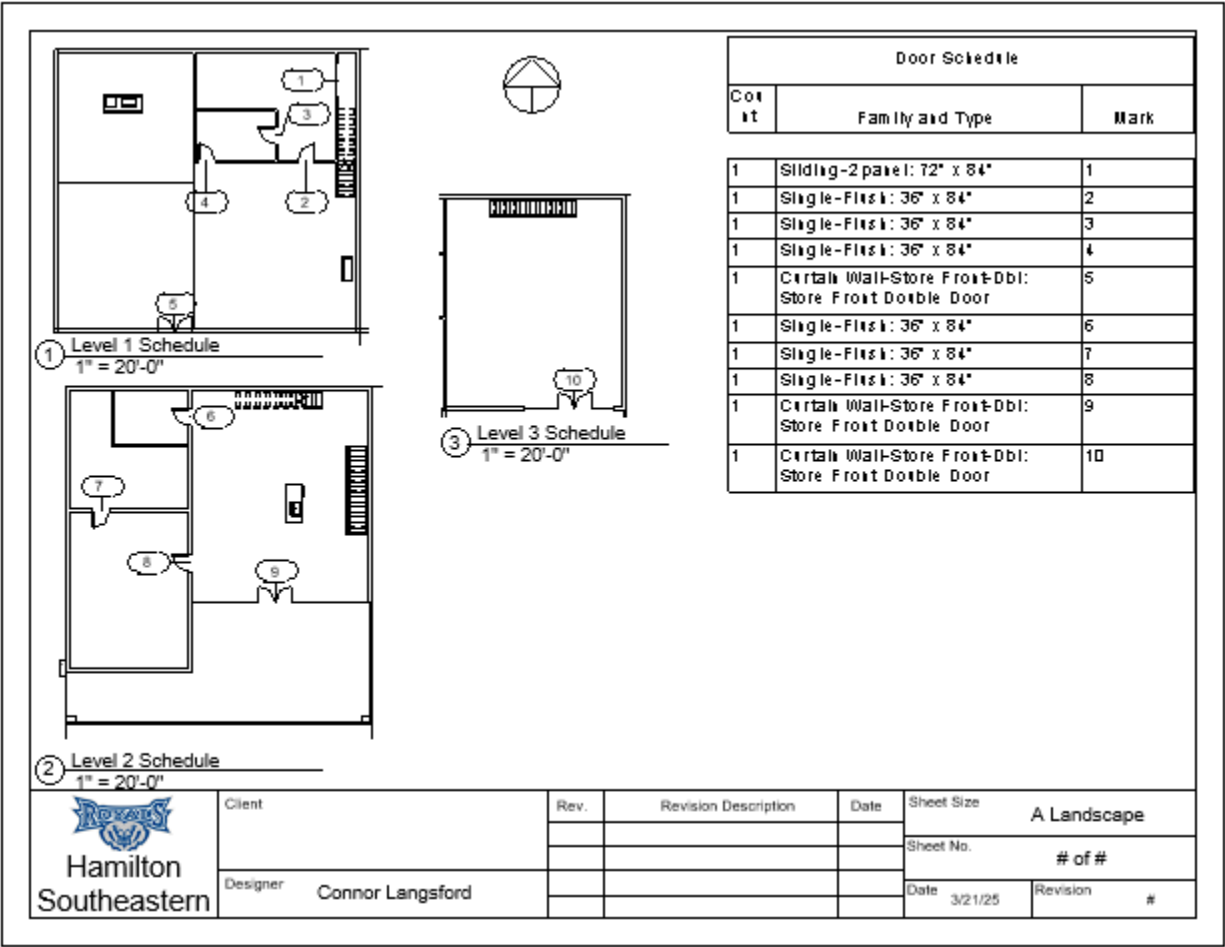
 Hamilton Southeastern	Client	Rev.	Revision Description	Date	Sheet Size	A Landscape
					Sheet No.	# of #
	Designer	Connor Langsford			Date	3/21/25
					Revision	#

Electrical Plan



Schedule

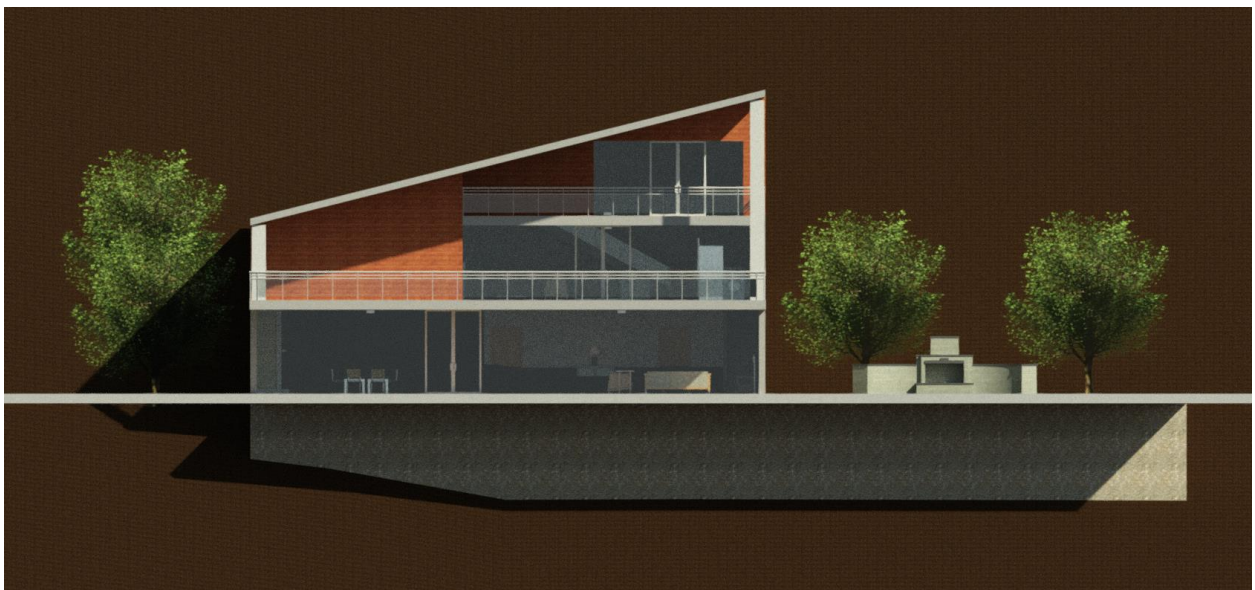
-Door Schedule



-Window Schedule

No windows only curtain walls

Renderings



Water Supply

-Property Facts

- Water line is 11 years old, 10in diameter, 5.25miles long, and cast iron
- The pipe has four 90° elbows, and four 45° elbows
- Located 9ft below grade
- 110gpm flow rate
- 843ft site elevation
- Lowest point of water tower is 1014ft

-Dynamic Head

dynamic head = static head – head loss

static head = Δ elevation

static head = 1014ft – 843ft

static head = 171ft

$$h_f = \frac{10.44 \times L \times Q^{1.85}}{C^{1.85} \times d^{4.8655}}$$

$$h_f = \frac{10.44 \left(5.25 \text{ miles} \times \frac{5280 \text{ ft}}{1 \text{ mile}} + 92 \text{ ft} \right) (110 \text{ gpm})^{1.85}}{(100)^{1.85} \times (10 \text{ in})^{4.8655}}$$

$$h_f = 4.721 \text{ ft}$$

dynamic head = 171ft – 4.721ft

dynamic head = 166.279ft

-Actual Water Pressure at the Water Meter

$$166.279 \text{ ft} \times \frac{1 \text{ psi}}{2.31 \text{ ft}}$$

Actual water pressure = 72psi

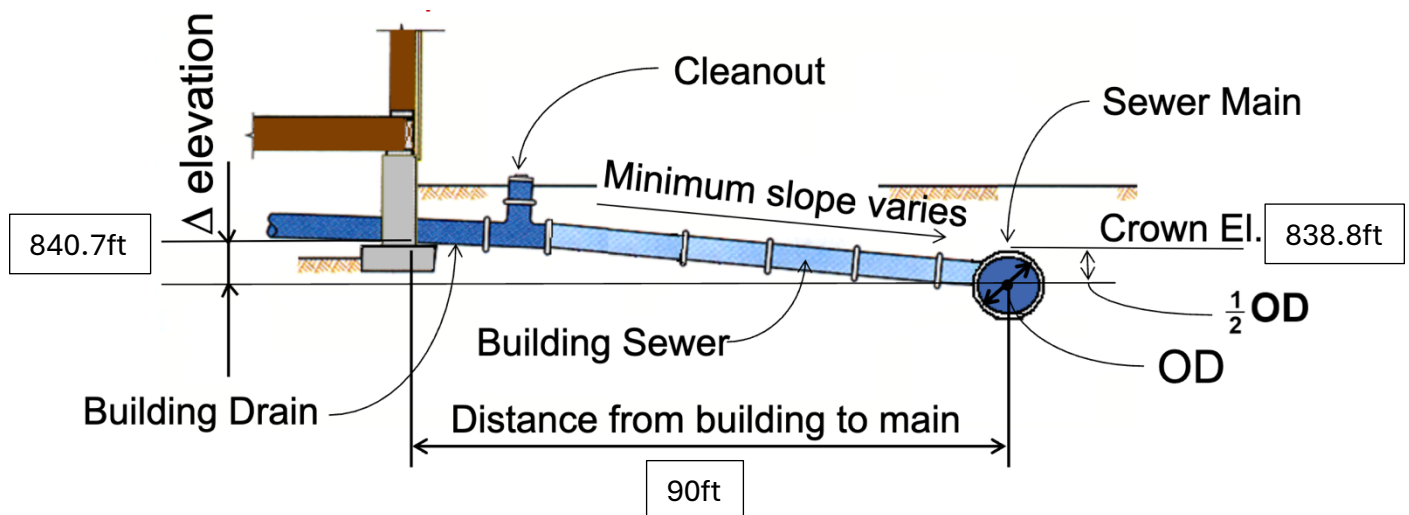
Residential water pressure should be between 40 – 80psi

Flanged Fittings		Pipe Size								
		1/2	3/4	6	8	10	12	14	16	18
Elbows	Regular 90 degree	0.9	1.2	8.9	12.0	14.0	17.0	18.0	21.0	23.0
	Long radius 90 degree	1.1	1.3	5.7	7.0	8.6	9.0	9.4	10.0	11.0
	Regular 45 degree	0.5	0.6	5.6	7.7	9.0	11.0	13.0	15.0	16.0
Tees	Line Flow	0.7	0.8	3.8	4.7	5.2	6.0	6.4	7.2	7.6
	Branch Flow	2.0	2.6	18.0	24.0	30.0	34.0	37.0	43.0	47.0
Return Bends	Regular 180 degree	0.9	1.2	8.9	12.0	14.0	17.0	18.0	21.0	23.0
	Long radius 180 degree	1.1	1.3	5.7	7.0	8.0	9.0	9.4	10.0	11.0
Valves	Globe	38.0	40.0	190.0	260.0	310.0	390.0			
	Gate			3.2	3.2	3.2	3.2	3.2	3.2	3.2
	Angle	15.0	15.0	63.0	90.0	120.0	140.0	160.0	190.0	210.0
	Swing Check	3.8	5.3	63.0	90.0	120.0	140.0			

Wastewater Discharge

-Property Facts

- 843ft grade elevation
- 16in sewer main
- 837.5ft invert elevation
- 20d.f.u drainage fixtures
- 3in pipe
- 25in frost depth
- 2% minimum allowable slope



-Crown Elevation

crown elevation = invert elevation + pipe diameter

$$\text{crown elevation} = 837.5\text{ft} + 16\text{in} \times \frac{1\text{ft}}{12\text{in}}$$

$$\text{crown elevation} = 838.833\text{ft}$$

-Max Invert Elevation

max invert elevation = grade elevation – frost depth – pipe diameter

$$\text{max invert elevation} = 843\text{ft} - 25\text{in} \times \frac{1\text{ft}}{12\text{in}} - 3\text{in} \times \frac{1\text{ft}}{12\text{in}}$$

$$\text{max invert elevation} = 840.667\text{ft}$$

-Sewer Lateral Slope

$$\text{sewer lateral slope} = \frac{\text{invert elevation at building} - \text{crown elevation at main} + \frac{1}{2}\text{OD}}{\text{distance from building to sewer main}}$$

$$\text{sewer lateral slope} = \frac{843\text{ft} - 838.833\text{ft} + \left(\frac{1}{2} \left(16\text{in} \times \frac{1\text{ft}}{12\text{in}}\right)\right)}{90\text{ft}} \times 100\%$$

$$\text{sewer lateral slope} = 5.4\%$$

Design meets min slope requirements because $5.4\% > 2\%$

Storm Water Runoff**-Property Facts**

- Farmland
- Site has well drained, sandy soil, 6.5% slope
- The design storm is a 10-yr 1-hr storm

-storm water runoff pre-development

$$Q = C_f \times C \times i \times A$$

$$Q = (1.0)\left(\frac{0.05+0.3}{2}\right)(2.17 \frac{in}{hr})(24265 ft^2 \times \frac{1 acre}{43,560 ft^2})$$

$$Q = 0.212 cfs$$

-storm water runoff post-development

$$C_c = \frac{C_1 A_1 + C_2 A_2 + C_3 A_3}{A_1 + A_2 + A_3}$$

$$C_c = \frac{(0.175)(0.408 acres) + (0.95)(0.075 acres) + (0.2)(0.075 acres)}{0.557 acres}$$

$$C_c = 0.28$$

$$Q_{post} = C_f \times C_c \times i \times A$$

$$Q_{post} = (1.0)(0.28)(2.17 \frac{in}{hr})(24265 ft^2 \times \frac{1 acre}{43,560 ft^2})$$

$$Q_{post} = 0.34 cfs$$

Return Period	C _r
1, 2, 5, 10	1.0
25	1.1
50	1.2
100	1.25

PDS-based precipitation frequency estimates with 90% confidence interval							
Duration	Average recurrence interval (years)						
	1	2	5	10	25	50	100
5-min	4.57 (4.22-4.96)	5.35 (4.97-5.82)	6.17 (5.69-6.71)	6.82 (6.28-7.40)	7.63 (7.00-8.29)	8.24 (7.50-8.95)	8.83 (7.99-9.67)
10-min	3.65 (3.38-3.95)	4.28 (3.97-4.66)	4.94 (4.55-5.37)	5.45 (5.02-5.92)	6.08 (5.57-6.61)	6.56 (5.98-7.12)	7.02 (6.35-7.69)
15-min	3.04 (2.82-3.30)	3.59 (3.32-3.90)	4.16 (3.84-4.53)	4.60 (4.23-4.99)	5.14 (4.71-5.58)	5.54 (5.04-6.01)	5.92 (5.38-6.46)
30-min	2.08 (1.93-2.26)	2.48 (2.30-2.69)	2.96 (2.73-3.22)	3.33 (3.07-3.62)	3.81 (3.49-4.14)	4.17 (3.80-4.53)	4.53 (4.10-4.96)
60-min	1.30 (1.20-1.41)	1.56 (1.44-1.69)	1.90 (1.75-2.06)	2.17 (2.00-2.36)	2.54 (2.32-2.76)	2.83 (2.57-3.07)	3.12 (2.82-3.42)
2-hr	0.768 (0.711-0.833)	0.914 (0.847-0.994)	1.11 (1.03-1.21)	1.27 (1.17-1.38)	1.48 (1.36-1.61)	1.66 (1.51-1.80)	1.83 (1.65-1.99)
3-hr	0.556 (0.514-0.605)	0.662 (0.612-0.722)	0.803 (0.742-0.873)	0.919 (0.845-1.00)	1.08 (0.984-1.17)	1.20 (1.09-1.31)	1.33 (1.20-1.44)
6-hr	0.335 (0.307-0.368)	0.398 (0.365-0.438)	0.481 (0.442-0.530)	0.553 (0.505-0.607)	0.650 (0.596-0.715)	0.731 (0.657-0.803)	0.816 (0.727-0.904)

Ground Cover	Runoff Coefficient, c
Lawns	0.05 - 0.35
Forest	0.05 - 0.25
Cultivated land	0.08-0.41
Meadow	0.1 - 0.5
Parks, cemeteries	0.1 - 0.25
Unimproved areas	0.1 - 0.3
Pasture	0.12 - 0.62
Residential areas	0.3 - 0.75
Business areas	0.5 - 0.95
Industrial areas	0.5 - 0.9
Asphalt streets	0.7 - 0.95
Brick streets	0.7 - 0.85
Roofs	0.75 - 0.95
Concrete streets	0.7 - 0.95
Lawns	0.05 - 0.35

Source: Lindsley, 1992

Heat Loss/Gain

Model

-Scale

1':54'

References

Frank Lloyd Wright Foundation. (2019). *About Frank Lloyd Wright*. Frank Lloyd Wright Foundation; Frank Lloyd Wright Foundation. <https://franklloydwright.org/frank-lloyd-wright/>